

## EAST WEST UNIVERSITY

Department of Computer Science and Engineering  
CSE475 — Machine Learning, Summer 2026

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### Report 3: Ablation, Final Model & Explainability

ds004504 EEG Dataset — Alzheimer's, Frontotemporal Dementia and Healthy Controls

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**Dataset:** OpenNeuro ds004504 – EEG in Alzheimer's Disease,  
Frontotemporal Dementia and Healthy Controls

**Track:** 1 – Graph Neural Network on Table-Style EEG Data

#### Submitted by – Group 03

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This report accompanies the notebooks `task3_improvement_ablation.ipynb` and

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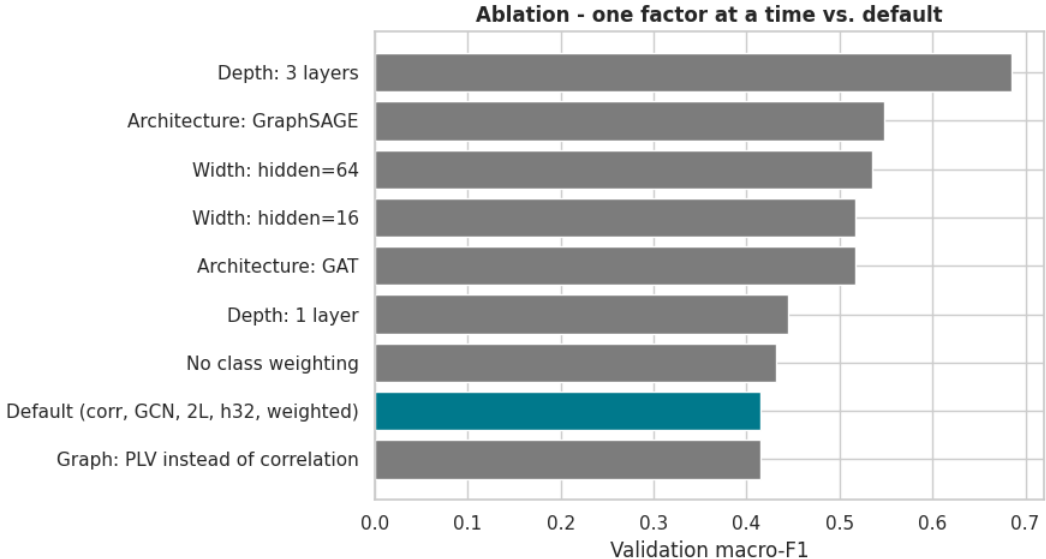
`task3_explainability.ipynb.`

## 1. Ablation Study

Starting from the Task 2b default configuration (correlation graph, 2-layer GCN, hidden size 32, class-weighted loss), one factor at a time was varied: graph type, GNN architecture (GAT, GraphSAGE), network depth, network width, and class weighting. Validation macro-F1 was recorded for each configuration, with the test set held out untouched.

| Configuration                          | Val. Macro-F1 |
|----------------------------------------|---------------|
| Depth: 3 layers                        | <b>0.686</b>  |
| Architecture: GraphSAGE                | 0.548         |
| Width: hidden = 64                     | 0.536         |
| Width: hidden = 16                     | 0.517         |
| Architecture: GAT                      | 0.517         |
| Depth: 1 layer                         | 0.444         |
| No class weighting                     | 0.432         |
| Default (corr, GCN, 2L, h32, weighted) | 0.415         |
| Graph: PLV instead of correlation      | 0.415         |

**Table 1:** One-factor-at-a-time ablation results vs. the Task 2b default.



**Figure 1:** Ablation results: validation macro-F1 per configuration.

The best configuration simply increases depth to 3 GCN layers (val. macro-F1 = 0.686), which was carried forward as the final architecture.

## 2. Final Model: 5-Fold Cross-Validation vs. Best Baseline

The best-ablation GNN (3-layer GCN) was compared against the best Task 2a baseline (SVM-RBF) using a leakage-safe, subject-grouped, stratified 5-fold cross-validation — each fold retrains both models from scratch on that fold’s training subjects only.

| Fold         | 1     | 2     | 3     | 4     | 5     |
|--------------|-------|-------|-------|-------|-------|
| SVM macro-F1 | 0.423 | 0.452 | 0.399 | 0.637 | 0.524 |
| GNN macro-F1 | 0.505 | 0.493 | 0.497 | 0.597 | 0.561 |

**Table 2:** Per-fold macro-F1: SVM-RBF vs. final GNN (3-layer GCN).

**SVM 5-fold macro-F1:**  $0.487 \pm 0.086$     **GNN 5-fold macro-F1:**  $0.531 \pm 0.041$

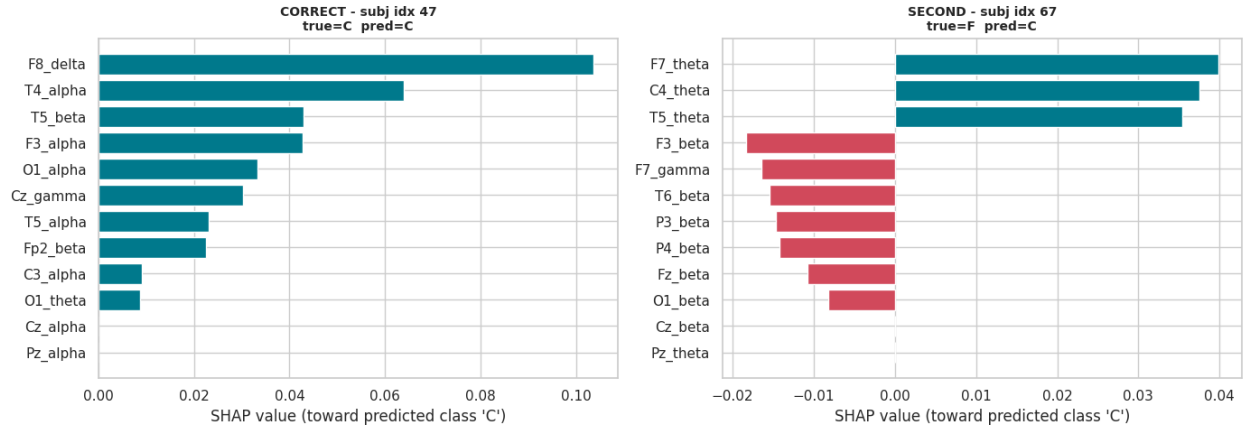
A paired Wilcoxon signed-rank test gave statistic = 2.000,  $p = 0.1875$  – **not** statistically significant at  $\alpha = 0.05$ , though the small number of folds ( $n = 5$ ) limits statistical power.

| Model                                             | Macro-F1     |
|---------------------------------------------------|--------------|
| Best Task 2a baseline (SVM-RBF, single split)     | 0.595        |
| First GNN (Task 2b, corr-graph GCN, single split) | 0.458        |
| Best-ablation GNN (3 layers, single split)        | 0.686        |
| SVM, 5-fold CV mean                               | 0.487        |
| Best-ablation GNN, 5-fold CV mean                 | <b>0.531</b> |
| Liu et al. (2026) DT-GCN, ds004504, 5-fold CV     | 0.790        |

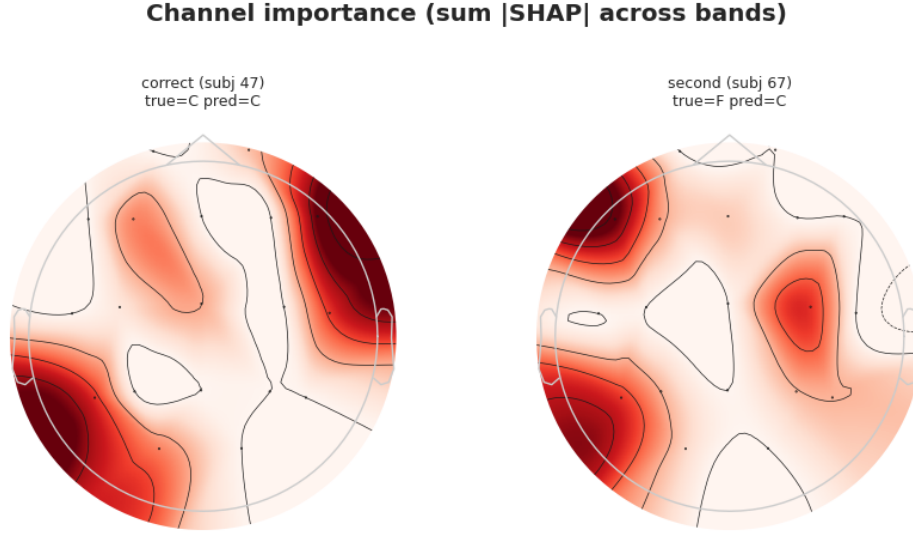
**Table 3:** Final comparison table. The Liu et al. reference is from a different train/test split in a different paper and is not a like-for-like statistical comparison.

### 3. Explainability (SHAP + LIME)

The final GNN’s predictions were explained for one correctly-classified subject and one misclassified subject, using SHAP (`KernelExplainer`) and LIME (`LimeTabularExplainer`) over the flattened 95-dimensional (19 channel  $\times$  5 band) node-feature vector. Each subject’s own connectivity graph is held fixed while SHAP/LIME perturb only the node features, so explanations answer “*given this subject’s connectivity pattern, which channel/band values drove the prediction?*”



**Figure 2:** Top-12 SHAP feature contributions for a correctly-classified subject (idx 47, true = CN, predicted = CN) and a misclassified subject (idx 67, true = FTD, predicted = CN).



**Figure 3:** Spatial topomap of per-channel importance (sum of |SHAP| across all 5 bands), plotted on the standard 10–20 montage, for the same two subjects.

### Reading

Agreement between SHAP and LIME on the dominant channel  $\times$  band features is a useful sanity check that these features genuinely drive the GNN rather than being an artifact of one explainer. Comparing the correct and misclassified panels shows that different, and partly conflicting, channel/band evidence underlies the two predictions — consistent with AD/FTD being a genuinely harder discrimination sub-problem, as already flagged in the EDA report. The topomap adds spatial context beyond the raw SHAP bar charts, showing that important channels are not uniformly distributed across the scalp but cluster in particular regions, a testable claim against the EEG/dementia literature.

### Caveat

These explanations are conditional on each subject’s own fixed connectivity graph — they attribute the prediction to channel-band values *given* that subject’s connectivity pattern, not to the connectivity pattern itself. Explaining which edges (not just which nodes) mattered (e.g. via GNNExplainer) is left for future work.

## 4. Summary

The ablation identified a simple depth increase (3-layer GCN) as the strongest single change over the Task 2b default. The resulting final model modestly outperforms the best non-graph baseline in 5-fold cross-validation (0.531 vs. 0.487 macro-F1), though the difference is not statistically significant at this sample size. SHAP/LIME explanations show the model relies on physiologically interpretable channel  $\times$  band features, with spatial patterns that differ between correctly and incorrectly classified subjects.